STA 445 S26 Assignment 5

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```r
library(tidyverse)
library(lubridate)
```

Problem 1

Convert the following to date or date/time objects.

a. September 13, 2010.

```r
mdy("September 13, 2010")
```

```
## [1] "2010-09-13"
```

b. Sept 13, 2010.

```r
mdy("Sept 13, 2010")
```

```
## Warning: All formats failed to parse. No formats found.
```

```
## [1] NA
```

Here I get a warning because "Sept" isn't an appropriate month for the mdy function.

c. Sep 13, 2010.

```r
mdy("Sep 13, 2010")
```

```
## [1] "2010-09-13"
```

d. S 13, 2010. Comment on the month abbreviation needs.

```r
mdy("S 13, 2010")
```

```
## Warning: All formats failed to parse. No formats found.
```

```
## [1] NA
```

Here I get a warning because "S" isn't an appropriate month for the mdy function, similar to Sept..

e. 07-Dec-1941.

```r
dmy("07-Dec-1941")
```

```
## [1] "1941-12-07"
```

f. 1-5-1998. Comment on why you might be wrong.

```r
mdy("1-5-1998")
```

```
## [1] "1998-01-05"
```

I might be wrong here because 01 could be the month or the day, and same thing with 05, it could also be the month or the day.

g. 21-5-1998. Comment on why you know you are correct.

```r
dmy("21-5-1998")
```

```
## [1] "1998-05-21"
```

I know am I right here because the only possibiliy for month is 05, meaning 21 has to be the day.

h. 2020-May-5 10:30 am

```r
ymd_hm("2020-May-5 10:30 am")
```

```
## [1] "2020-05-05 10:30:00 UTC"
```

i. 2020-May-5 10:30 am PDT (ex Seattle)

```r
ymd_hm("2020-May-5 10:30 am", tz="America/Vancouver")
```

```
## [1] "2020-05-05 10:30:00 PDT"
```

j. 2020-May-5 10:30 am AST (ex Puerto Rico)

```r
ymd_hm("2020-May-5 10:30 am", tz="America/Puerto_Rico")
```

```
## [1] "2020-05-05 10:30:00 AST"
```

Problem 2

Using just your date of birth (ex Sep 7, 1998) and today's date calculate the following:

```r
tate_birthday <- mdy("Mar 24, 2004")
```

a. Calculate the date of your 64th birthday.

```r
(tate_birthday + years(64))
```

```
## [1] "2068-03-24"
```

b. Calculate your current age (in years).

```r
current_age <- year(as.period(interval(tate_birthday, Sys.Date())))
current_age
```

```
## [1] 21
```

c. Using your result in part (b), calculate the date of your next birthday.

```r
next_birthday <- years(current_age) + tate_birthday + years(1)
next_birthday
```

```
## [1] "2026-03-24"
```

d. The number of *days* until your next birthday.

```r
day_interval <- interval(Sys.Date(), next_birthday)
as.period(day_interval, units = "days")
```

```
## [1] "8d 0H 0M 0S"
```

e. The number of *months* and *days* until your next birthday.

```r
as.period(day_interval, units = "months")
```

```
## [1] "8d 0H 0M 0S"
```

Problem 3

Suppose you have arranged for a phone call to be at 3 pm on May 8, 2015 at Arizona time. However, the recipient will be in Auckland, NZ. What time will it be there?

```r
az_time <- mdy_h("May 8, 2015 3 pm", tz = "US/Arizona")
```

```r
with_tz(az_time, tz="Pacific/Auckland")
```

```
## [1] "2015-05-09 10:00:00 NZST"
```

The time in Auckland, NZ would be 10:00:00 on May 9th, 2015.

Problem 4

It turns out there is some interesting periodicity regarding the number of births on particular days of the year.

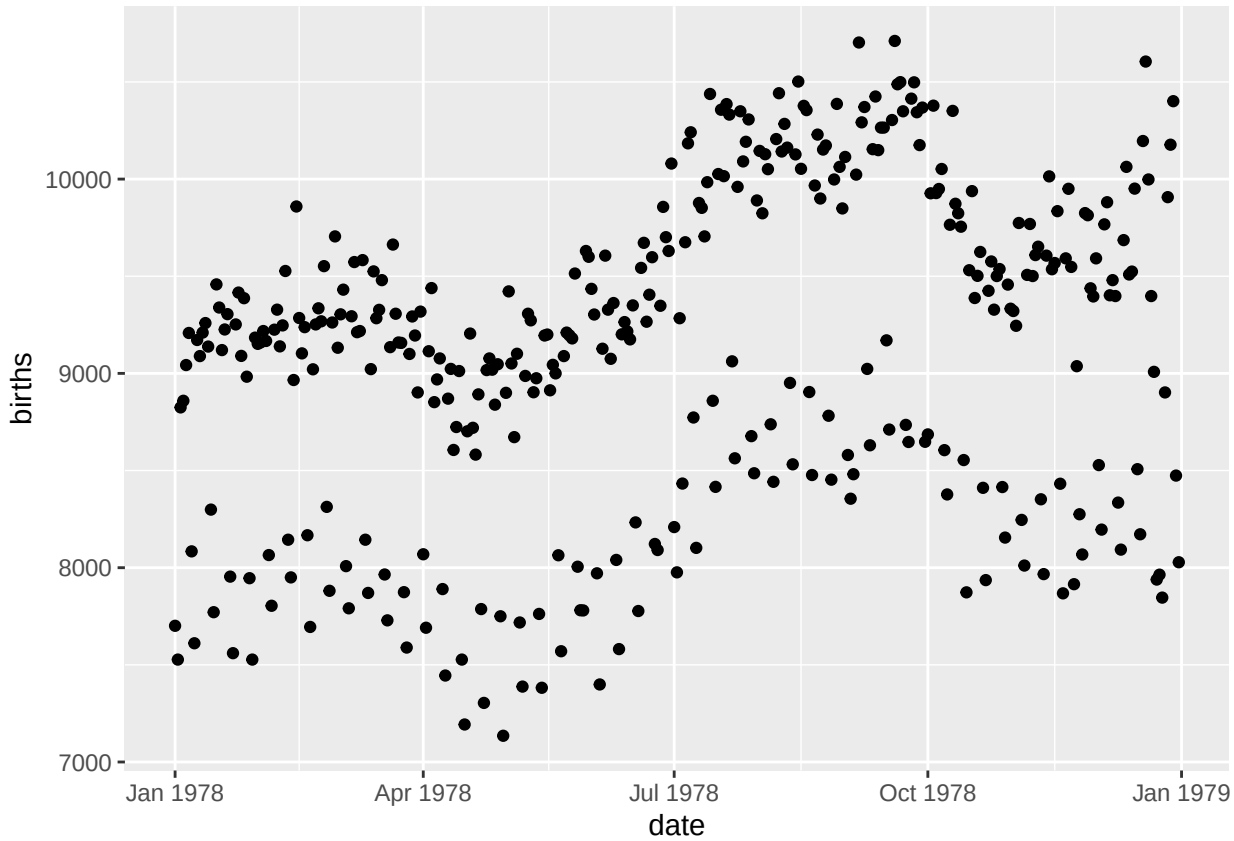
a. Using the `mosaicData` package, load the data set `Births78` which records the number of children born on each day in the United States in 1978. Because this problem is intended to show how to calculate the information using the `date`, remove all the columns *except* `date` and `births`.

```r
library(mosaicData)
data(Births78)
Births78 <- Births78 %>%
  select(date, births)
head(Births78)
```

```
##         date births
## 1 1978-01-01   7701
## 2 1978-01-02   7527
## 3 1978-01-03   8825
## 4 1978-01-04   8859
## 5 1978-01-05   9043
## 6 1978-01-06   9208
```

b. Graph the number of `births` vs the `date` with date on the x-axis. What stands out to you? Why do you think we have this trend?

```r
births_graph <- ggplot( Births78, aes(x = date, y = births) ) +
  geom_point()
births_graph
```



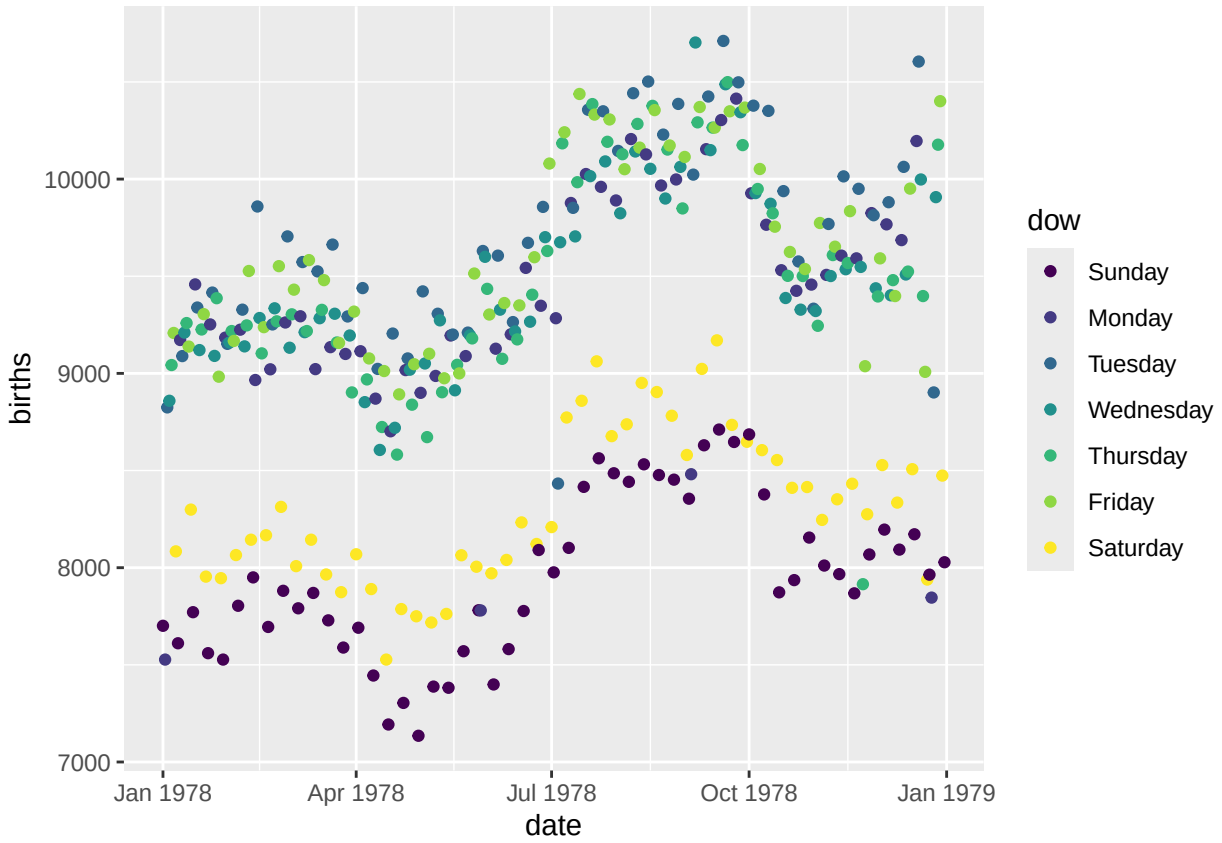
The graph points follow a wave pattern, with high births in January of 1978, then a dip in the spring of 1978, followed by an increase in the fall and winter of 1978. This could be because of weather, geo-political reasons, or holidays.

c. To test your assumption, we need to figure out the what day of the week each observation is. Use `dplyr::mutate` to add a new column named `dow` that is the day of the week (Monday, Tuesday, etc). This calculation will involve some function in the `lubridate` package and the `date` column.

```
Births78 <- Births78 %>%
  mutate(
    dow = wday( date, label = TRUE, abbr = FALSE )
  )
```

d. Plot the data with the point color being determined by the day of the week variable.

```
births_weeks_graph <- ggplot( Births78, aes(x = date, y = births) ) +
  geom_point(aes( color = dow ) )
births_weeks_graph
```


10000
9000
8000
7000
births
Jan 1978
Apr 1978
Jul 1978
Oct 1978
Jan 1979
date
dow
Sunday
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday